EXPLICIT POISSON–HADAMARD ZERO-FREE REGION INEQUALITIES, QUASICONCAVE MINORANT CROSSING, RIGOROUS TRUDGIAN STIELTJES BOUNDS, AND COMPARISON WITH SMOOTH WEIL WEIGHTS

THIJS MISSIAEN

Abstract. We examine explicit de la Vallée Poussin and Stechkin zero-free region inequalities for $\zeta(s)$ through the mirror-paired Hadamard Poisson kernel $K(d,\delta,u) = \frac{1}{2}[\frac{d-\delta}{(d-\delta)^2+u^2} + \frac{d+\delta}{(d+\delta)^2+u^2}]$ ($d > \frac{1}{2}$, $\delta = |\beta - \frac{1}{2}| \in [0, \frac{1}{2})$), connected to the canonical transverse Wronskian $W(d,t) = 16\int_0^\infty y\sinh(2dy)I(y,t)\,\mathrm{d}y$. While $\int_{\mathbb{R}} K(d,\delta,u)\,\mathrm{d}u = \pi$ is conserved, we prove that $K(d,\frac{1}{2},u)$ is not a global minorant on $\mathbb{R}$: because $\delta^2 \mapsto K(d,\delta,u)$ is strictly quasiconcave on $[0,\frac{1}{4}]$, the sharp pointwise minorant is $K_\flat(d,u) = \min(K(d,0,u), K(d,\frac{1}{2},u))$ with crossing at $u_\times(d) < d/\sqrt{3}$, validating plateau minorants $\widetilde{K}_H(d,u)$ precisely for $H \geq u_\times(d)$. Incorporating Trudgian's unconditional bound $|S(T)| \leq 0.112\ln T + 0.278\ln\ln T + 2.510$, logarithmic tail and concavity corrections, and focal-pair de-duplication at $u = 0$, we benchmark plateau subtraction against Mossinghoff–Trudgian–Yang ($R = 5.558691$) and explain why sharp Poisson truncation trails smooth Heath-Brown Weil weights.

1. Introduction and Main Results

Under the canonical normalization established in Papers V and VI [11, 12], Riemann's $\Xi(z) := \xi(\frac{1}{2}+\mathrm{i}z) = 8\int_0^\infty \Phi(u)\cos(2zu)\,\mathrm{d}u$ ($z = t - \mathrm{i}d$, $s = \frac{1}{2}+d+\mathrm{i}t$) [16, 7, 15] has transverse Wronskian [17, 10, 5, 1, 22, 6]

$$W(d,t) := \frac{1}{2}\frac{\partial}{\partial d}|\Xi(t-\mathrm{i}d)|^2 = |\Xi(t-\mathrm{i}d)|^2\,\mathrm{Re}\frac{\xi'(\frac{1}{2}+d+\mathrm{i}t)}{\xi(\frac{1}{2}+d+\mathrm{i}t)} = 16\int_0^\infty y\sinh(2dy)\,I(y,t)\,\mathrm{d}y. \quad (1.1)$$

For $d > \frac{1}{2}$ ($\sigma = \frac{1}{2}+d > 1$), $\Xi(t-\mathrm{i}d) \neq 0$, and dividing $W(d,\tau)$ by $|\Xi(\tau-\mathrm{i}d)|^2$ links Hadamard's zero-sum partial fraction $\mathrm{Re}(\xi'/\xi)(s) = \sum_\rho \mathrm{Re}(s-\rho)^{-1}$ directly to the Dirichlet series $-\mathrm{Re}(\zeta'/\zeta)(s) = \sum_{n=2}^\infty \Lambda(n)n^{-1/2-d}\cos(\tau\ln n)$ via $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$.

Classical explicit zero-free regions—initiated by de la Vallée Poussin [2] and sharpened via functional-equation zero pairing by Stechkin [18], Kondratiev [8], Heath-Brown [4], and Mossinghoff, Trudgian, and Yang [13]—combine non-negative trigonometric polynomials $P(\theta) = \frac{1}{2}+\sum_{k=1}^K w_k\cos(k\theta) \geq 0$ with lower bounds on the background nontrivial zero sum. Meanwhile, Platt and Trudgian [14] verified unconditionally that every nontrivial zero with $|\gamma| \leq 3\cdot 10^{12}$ lies on $\sigma = \frac{1}{2}$, while Mossinghoff, Trudgian, and Yang [13] established the classical region $\sigma \geq 1 - (5.558691\ln|t|)^{-1}$ ($|t| \geq 2$) and the asymptotically wider Korobov–Vinogradov [9, 21] region $\sigma \geq 1 - [55.241(\ln|t|)^{2/3}(\ln\ln|t|)^{1/3}]^{-1}$ ($|t| \geq 3$).

This methodological note clarifies the exact algebra and analytical pitfalls of plateau-truncated Poisson–Hadamard background subtraction:

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Theorem 1.1 (Quasiconcave Minorant Crossing and Rigorous Plateau Stieltjes Bound)**.** *Let* $d > \frac{1}{2}$, $\delta \in [0, \frac{1}{2}]$, *and* $K(d,\delta,u) := \frac{1}{2}[\frac{d-\delta}{(d-\delta)^2+u^2} + \frac{d+\delta}{(d+\delta)^2+u^2}]$.

(i) ***Toeplitz Weight Bound and Stechkin Kernel Conservation.*** *Every degree-*K *non-negative cosine polynomial* $P(\theta) = \frac{1}{2} + \sum_{k=1}^{K} w_k \cos(k\theta) \geq 0$ *satisfies* $w_1 \leq \cos(\frac{\pi}{K+2})$ [3, 19], *with extremal sum* $\sum_{k=1}^{K} w_k^{(K)} = \frac{1}{K+2}\cot^2(\frac{\pi}{2(K+2)}) - \frac{1}{2}$. *For all* $0 \leq \delta < d$, $\int_{-\infty}^{\infty} K(d,\delta,u)\,\mathrm{d}u = \pi$.

(ii) ***Quasiconcave Endpoint Crossing.*** *Because* $\int_{\mathbb{R}}[K(d,\delta,u) - K(d,\frac{1}{2},u)]\,\mathrm{d}u = 0$, $K(d,\frac{1}{2},u)$ *is* not *a minorant near* $u = 0$. *Instead,* $\delta^2 \mapsto K(d,\delta,u)$ *is strictly quasiconcave on* $[0,\frac{1}{4}]$, *so* $\min_{\delta\in[0,1/2]} K(d,\delta,u) = \min(K(d,0,u), K(d,\frac{1}{2},u))$, *crossing uniquely on* $(0,\infty)$ *at* $u_\times(d) < d/\sqrt{3}$ *in* (4.3). *Consequently,* $\widetilde{K}_H(d,u) := K(d,\frac{1}{2},\max(|u|,H))$ *is a valid global minorant on* $\mathbb{R}$ *whenever* $H \geq u_\times(d)$.

(iii) ***Rigorous Trudgian Stieltjes Lower Bound and Focal De-Duplication.*** *Using Trudgian's bound* [20] $|S(T)| \leq B_S^{(0)}(T) := 0.112\ln T + 0.278\ln\ln T + 2.510$ *(*$T \geq \mathrm{e}$*), every ordinate* $\tau \geq 30$ *and* $H \in [u_\times(d), \tau/2]$ *satisfies the unconditional bound* (5.2), *where extracting a putative off-line pair* $\{\rho_0, 1-\bar{\rho}_0\}$ *at* $\gamma = t$ *requires subtracting* $2K(d,\frac{1}{2},H)$ *at* $k = 1$ *to prevent* $u = 0$ *double counting. Because piecewise-smooth plateau corners incur Stieltjes boundary jumps* $\approx 2B_S^{(0)}(\tau)K(d,\frac{1}{2},H)$, *plateau Poisson subtraction yields* $R_{\mathrm{eff}} > 5.558691$, *trailing smooth compactly supported Fourier/Weil weights* [4, 13].

1.1. **Proof overview.** The proofs rely on three analytical steps:

(1) *Fejér–Riesz factorization and reflection bijection (Sections 2–3).* Writing $P(\theta) = \frac{1}{2}|\sum_{j=0}^{K} a_j \mathrm{e}^{\mathrm{i}j\theta}|^2$ reduces $\max w_1$ to the tridiagonal Toeplitz eigenvalue $2\cos(\frac{\pi}{K+2})$, while the involution $\rho \mapsto 1-\bar{\rho}$ pairs the multiset of nontrivial zeros into $K(d, d_\rho, \tau-\gamma)$ without miscounting on-line versus off-line multiplicities.

(2) *Biquadratic difference factorization and* δ^2*-monotonicity (Section 4).* Factoring $K(d,\delta,u) - K(d,\frac{1}{2},u)$ isolates $3u^4 + (2d^2 + \frac{1}{4} + \delta^2)u^2 - (d^2 - \frac{1}{4})(d^2 - \delta^2)$, while $\partial_{\delta^2} K(d,\delta,u)$ changes sign at most once from $+$ to $-$, forcing the minimum onto $\{\delta = 0, \delta = \frac{1}{2}\}$.

(3) *Stieltjes integration by parts with logarithmic tail and concavity control (Section 5).* Bounding $B_S^{(0)}(\tau+u) - B_S^{(0)}(\tau+H) \leq c_{\mathrm{tail}}(\tau+H)\ln(\frac{\tau+u}{\tau+H})$ closes the $[\tau+H,\infty)$ tail in elementary arctangents, while symmetric pairing on $[\tau - \frac{\tau}{2}, \tau + \frac{\tau}{2}]$ bounds concavity of $\ln(\gamma/2\pi)$ explicitly.

1.2. **Notation and canonical normalizations.** Throughout, $s = \sigma + \mathrm{i}\tau = \frac{1}{2} + d + \mathrm{i}\tau$ with $d > \frac{1}{2}$ ($\sigma > 1$), $\mathcal{A}_\Gamma(d,\tau) := \frac{1}{2}\mathrm{Re}\,\psi(\frac{1}{4} + \frac{d}{2} + \mathrm{i}\frac{\tau}{2}) - \frac{1}{2}\ln\pi$, and nontrivial zeros $\rho = \beta + \mathrm{i}\gamma$ have displacement $d_\rho := |\beta - \frac{1}{2}| \in [0, \frac{1}{2})$. Backlund's counting decomposition is $N(T) = \frac{1}{\pi}\vartheta(T) + 1 + S(T)$ for $T > 0$ ($N(14) = 0$ since $\gamma_1 \approx 14.134725$).

2. Fejér–Lukács Trigonometric Polynomials and Toeplitz Bounds

Proposition 2.1 (Fejér–Lukács Eigenvalue Bound and Weight Sum [3, 19, 8])**.** *Let* $P(\theta) = \frac{1}{2} + \sum_{k=1}^{K} w_k \cos(k\theta) \geq 0$ *for all* $\theta \in \mathbb{R}$ *(*$w_k \in \mathbb{R}$, $K \geq 1$*). Then there exist* $a_0, \ldots, a_K \in \mathbb{R}$ *with* $\sum_{j=0}^{K} a_j^2 = 1$ *and* $w_k = \sum_{j=0}^{K-k} a_j a_{j+k}$. *Consequently,*

$$w_1 \leq \frac{1}{2}\Lambda_{\max}(K) = \cos\left(\frac{\pi}{K+2}\right), \tag{2.1}$$

attained uniquely (for $a_j \ge 0$) by $a_j^{(K)} := \sqrt{\frac{2}{K+2}} \sin \frac{(j+1)\pi}{K+2}$, for which

$$\sum_{k=1}^{K} w_k^{(K)} = \frac{1}{K+2} \cot^2\!\left(\frac{\pi}{2(K+2)}\right) - \frac{1}{2}. \tag{2.2}$$

Proof. By the Fejér–Riesz theorem [19, Thm. 1.2.1], $2P(\theta) = |\sum_{j=0}^{K} a_j e^{ij\theta}|^2 = \sum_{j=0}^{K} a_j^2 + 2\sum_{k=1}^{K}(\sum_{j=0}^{K-k} a_j a_{j+k}) \cos(k\theta)$, giving $\sum a_j^2 = 1$ and $2w_1 = a^{\mathsf{T}} J_K a$, where J_K is the $(K+1) \times (K+1)$ tridiagonal shift matrix with eigenvalues $2\cos\frac{m\pi}{K+2}$ $(1 \le m \le K+1)$, proving (2.1). Evaluating $P(0) = \frac{1}{2} + \sum_{k=1}^{K} w_k^{(K)} = \frac{1}{2}(\sum_{j=0}^{K} a_j^{(K)})^2$ via $\sum_{j=0}^{K} \sin\frac{(j+1)\pi}{K+2} = \cot\frac{\pi}{2(K+2)}$ yields (2.2). □

3. The Stechkin–Hadamard Mirror-Paired Kernel

Proposition 3.1 (Mirror-Paired Partial-Fraction Identity [18])**.** *Let $d > \frac{1}{2}$ and $s = \frac{1}{2} + d + i\tau$. Over the multiset of all nontrivial zeros $\rho = \beta + i\gamma$ ($d_\rho = |\beta - \frac{1}{2}| \in [0, \frac{1}{2})$),*

$$-\mathrm{Re}\frac{\zeta'(s)}{\zeta(s)} = \mathrm{Re}\frac{1}{s-1} + \mathcal{A}_\Gamma(d,\tau) - \sum_{\rho} K(d, d_\rho, \tau - \gamma), \tag{3.1}$$

where for $0 \le \delta < d$ and $u \in \mathbb{R}$ the symmetric Poisson–Hadamard kernel satisfies

$$\boxed{K(d,\delta,u) := \frac{1}{2}\left[\frac{d-\delta}{(d-\delta)^2+u^2} + \frac{d+\delta}{(d+\delta)^2+u^2}\right] = \frac{d(d^2-\delta^2+u^2)}{(d^2-\delta^2+u^2)^2 + 4u^2\delta^2} > 0,} \tag{3.2}$$

with conserved total integral $\int_{-\infty}^{\infty} K(d,\delta,u)\,du = \pi$. In particular, a putative off-line pair $\{\rho_0, 1-\bar{\rho}_0\}$ at $\rho_0 = \frac{1}{2} + d_0 + it$ ($d_0 \in (0, \frac{1}{2})$) consists of two distinct zeros in (3.1), contributing $2K(d, d_0, 0) = \frac{2d}{d^2 - d_0^2}$ at $\tau = t$.

Proof. Differentiating $\ln \xi(s) = \ln[\frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)]$ and cancelling the $s = 0$ pole gives $-\mathrm{Re}(\zeta'/\zeta)(s) = \mathrm{Re}(s-1)^{-1} + \mathcal{A}_\Gamma(d,\tau) - \sum_\rho \mathrm{Re}(s-\rho)^{-1}$. Because the involution $\rho \mapsto 1 - \bar{\rho}$ permutes the multiset of nontrivial zeros while preserving $\gamma = \mathrm{Im}(\rho)$ and $d_\rho = |\beta - \frac{1}{2}|$, averaging $\mathrm{Re}(s-\rho)^{-1}$ with $\mathrm{Re}(s-(1-\bar{\rho}))^{-1}$ gives $K(d, d_\rho, \tau - \gamma)$ term-by-term. Combining the two fractions in (3.2) via $(d-\delta)[(d+\delta)^2+u^2] + (d+\delta)[(d-\delta)^2+u^2] = 2d(d^2-\delta^2+u^2)$ and $((d-\delta)^2+u^2)((d+\delta)^2+u^2) = (d^2-\delta^2+u^2)^2 + 4u^2\delta^2$ proves the quartic form, while $\int_{\mathbb{R}} \frac{a}{a^2+u^2}\,du = \pi$ for $a = d \pm \delta > 0$ proves $\int_{\mathbb{R}} K\,du = \pi$. □

4. Quasiconcave Minorant Crossing and Plateau Admissibility

Because $\int_{\mathbb{R}} K(d,\delta,u)\,du = \pi$ is independent of $\delta \in [0, \frac{1}{2}]$, two kernels $K(d,\delta,u)$ and $K(d,\frac{1}{2},u)$ with $\delta < \frac{1}{2}$ cannot satisfy $K(d,\delta,u) \ge K(d,\frac{1}{2},u)$ for all $u \in \mathbb{R}$: at $u = 0$, $K(d,0,0) = \frac{1}{d} < \frac{d}{d^2 - 1/4} = K(d,\frac{1}{2},0)$. The exact pointwise envelope is governed by strict quasiconcavity in δ^2.

Theorem 4.1 (Biquadratic Crossing and Sharp Quasiconcave Minorant)**.** *Let $d > \frac{1}{2}$, $\delta \in [0, \frac{1}{2})$, and $D(d,\delta,u) := (d^2-\delta^2+u^2)^2 + 4u^2\delta^2 > 0$.*

(i) ***Exact Difference and Derivative Identities.*** *For all $u \in \mathbb{R}$,*

$$K(d,\delta,u) - K\!\left(d,\tfrac{1}{2},u\right) = \frac{d(\frac{1}{4}-\delta^2)\,[3u^4 + (2d^2 + \frac{1}{4} + \delta^2)u^2 - (d^2 - \frac{1}{4})(d^2 - \delta^2)]}{D(d,\delta,u)\,D(d,\frac{1}{2},u)}, \tag{4.1}$$

and for all $\delta \in [0, \frac{1}{2}]$,

$$\frac{\partial K(d,\delta,u)}{\partial(\delta^2)} = \frac{d\big[(d^2-\delta^2)^2 - 2(d^2+\delta^2)u^2 - 3u^4\big]}{D(d,\delta,u)^2}. \tag{4.2}$$

(ii) ***Sharp Endpoint Minorant and Crossing Radius.*** *For every fixed* (d,u), $\delta^2 \mapsto K(d,\delta,u)$ *is quasiconcave on* $[0,\frac{1}{4}]$ *and attains its minimum on* $\{0,\frac{1}{4}\}$, *so*

$$K_\flat(d,u) := \min_{\delta\in[0,1/2]} K(d,\delta,u) = \begin{cases} K(d,0,u) = \dfrac{d}{d^2+u^2}, & |u| \le u_\times(d), \\ K(d,\frac{1}{2},u), & |u| > u_\times(d), \end{cases} \tag{4.3}$$

where $u_\times(d) := \big[\frac{1}{6}\big(\sqrt{16d^4 - 2d^2 + \frac{1}{16}} - (2d^2+\frac{1}{4})\big)\big]^{1/2} < \frac{d}{\sqrt{3}}$.

(iii) ***Plateau Admissibility.*** *For every* $H \ge u_\times(d)$, $\widetilde{K}_H(d,u) := K(d,\frac{1}{2},\max(|u|,H))$ *satisfies* $0 < \widetilde{K}_H(d,u) \le K_\flat(d,u) \le K(d,\delta,u)$ *for all* $u \in \mathbb{R}$ *and all* $\delta \in [0,\frac{1}{2}]$.

Proof. (i) Writing $A_\delta = d^2 - \delta^2 + u^2$ and $A_{1/2} = d^2 - \frac{1}{4} + u^2$ ($A_\delta - A_{1/2} = \frac{1}{4} - \delta^2$), the numerator of $[K(d,\delta,u) - K(d,\frac{1}{2},u)]/d$ equals $A_\delta(A_{1/2}^2 + u^2) - A_{1/2}(A_\delta^2 + 4u^2\delta^2) = (\frac{1}{4} - \delta^2)[4u^2(d^2+u^2) - A_\delta A_{1/2}]$. Expanding $4u^2(d^2+u^2) - (u^2+d^2-\delta^2)(u^2+d^2-\frac{1}{4})$ gives (4.1), and letting $\delta_2^2 \to \delta_1^2$ yields (4.2).

(ii) In (4.2), as $x = \delta^2$ increases on $[0,\frac{1}{4}]$ ($x < d^2$), $(d^2-x)^2 - 2(d^2+x)u^2 - 3u^4$ decreases strictly with derivative $-2(d^2 - x + u^2) < 0$. Hence $\partial_x K$ can change sign at most once, from $+$ to $-$, proving quasiconcavity on $[0,\frac{1}{4}]$ and $\min_{\delta\in[0,1/2]} K(d,\delta,u) = \min(K(d,0,u), K(d,\frac{1}{2},u))$. Setting $\delta = 0$ in (4.1) shows $K(d,0,u) - K(d,\frac{1}{2},u)$ changes sign uniquely on $(0,\infty)$ at $3u^4 + (2d^2+\frac{1}{4})u^2 - d^2(d^2-\frac{1}{4}) = 0$, giving $u_\times(d)$. At $u^2 = d^2/3$, $3u^4 + (2d^2+\frac{1}{4})u^2 - d^2(d^2-\frac{1}{4}) = \frac{d^2}{3} > 0$, so $u_\times(d) < d/\sqrt{3}$.

(iii) Since $d - \frac{1}{2} > 0$, both $v \mapsto K(d,0,v)$ and $v \mapsto K(d,\frac{1}{2},v)$ decrease strictly on $[0,\infty)$, so $v \mapsto K_\flat(d,v)$ decreases strictly on $[0,\infty)$. For $H \ge u_\times(d)$ and $|u| \le H$, $K_\flat(d,u) \ge K_\flat(d,H) = K(d,\frac{1}{2},H) = \widetilde{K}_H(d,u)$, while for $|u| > H \ge u_\times(d)$, $K_\flat(d,u) = K(d,\frac{1}{2},u) = \widetilde{K}_H(d,u)$. □

5. Rigorous Trudgian Stieltjes Bound and Smooth-Weight Comparison

Trudgian [20, Thm. 1] proved unconditionally for all $T \ge \mathrm{e}$ that

$$|S(T)| \le B_S^{(0)}(T) := 0.112\ln T + 0.278\ln\ln T + 2.510. \tag{5.1}$$

In a constant-density toy model where $N'(\gamma)$ is frozen at $\frac{1}{2\pi}\ln(\tau/2\pi)$ on $\mathbb{R}$ and $|S(\gamma)|$ is frozen at $B_S^{(0)}(\tau)$, Stieltjes integration of $\widetilde{K}_H(d,\tau-\gamma)$ produces $\mathcal{F}_{\mathrm{model}}(d,\tau;H) := \frac{1}{2\pi}\ln\frac{\tau}{2\pi}[2HK_{\min}(d,H) + \pi - 2\Psi_{\mathrm{win}}(d,H)] - 2B_S^{(0)}(\tau)K_{\min}(d,H)$ ($K_{\min} := K(\cdot,\frac{1}{2},\cdot)$, $\Psi_{\mathrm{win}}(d,H) := \int_0^H K_{\min}(d,u)\,\mathrm{d}u$), whose H-derivative $[2H\cdot\frac{1}{2\pi}\ln(\tau/2\pi) - 2B_S^{(0)}(\tau)]\partial_H K_{\min}(d,H)$ vanishes at $H_{\mathrm{stat}}(\tau) := \frac{2\pi B_S^{(0)}(\tau)}{\ln(\tau/2\pi)}$ independently of d. Converting this heuristic into an unconditional inequality requires four rigorous corrections: (1) the true Trudgian constant 2.510 in (5.1); (2) logarithmic growth of $B_S^{(0)}(\gamma)$ on $[\tau+H,\infty)$; (3) support $N(14) = 0$ and concavity of $\ln(\gamma/2\pi)$; and (4) subtracting $2K_{\min}(d,H)$ at $k = 1$ so an extracted off-line pair $\{\rho_0, 1-\bar{\rho}_0\}$ at $\gamma = t$ is not counted twice.

Theorem 5.1 (Unconditional Plateau Background Lower Bound)**.** *Let* $d > \frac{1}{2}$, $\tau \ge 30$, $T_0 := 14$, $M := \frac{\tau}{2}$, *and* $u_\times(d) \le H \le M$. *Set* $\Phi_{\mathrm{plat}}(d,H;M) := 2HK_{\min}(d,H) + 2[\Psi_{\mathrm{win}}(d,M) - \Psi_{\mathrm{win}}(d,H)]$, $I_2(d,H;M) := \frac{1}{3}H^3K_{\min}(d,H) + \int_H^M u^2K_{\min}(d,u)\,\mathrm{d}u$, *and* $c_{\mathrm{tail}}(x) := 0.112 + \frac{0.278}{\ln x}$.

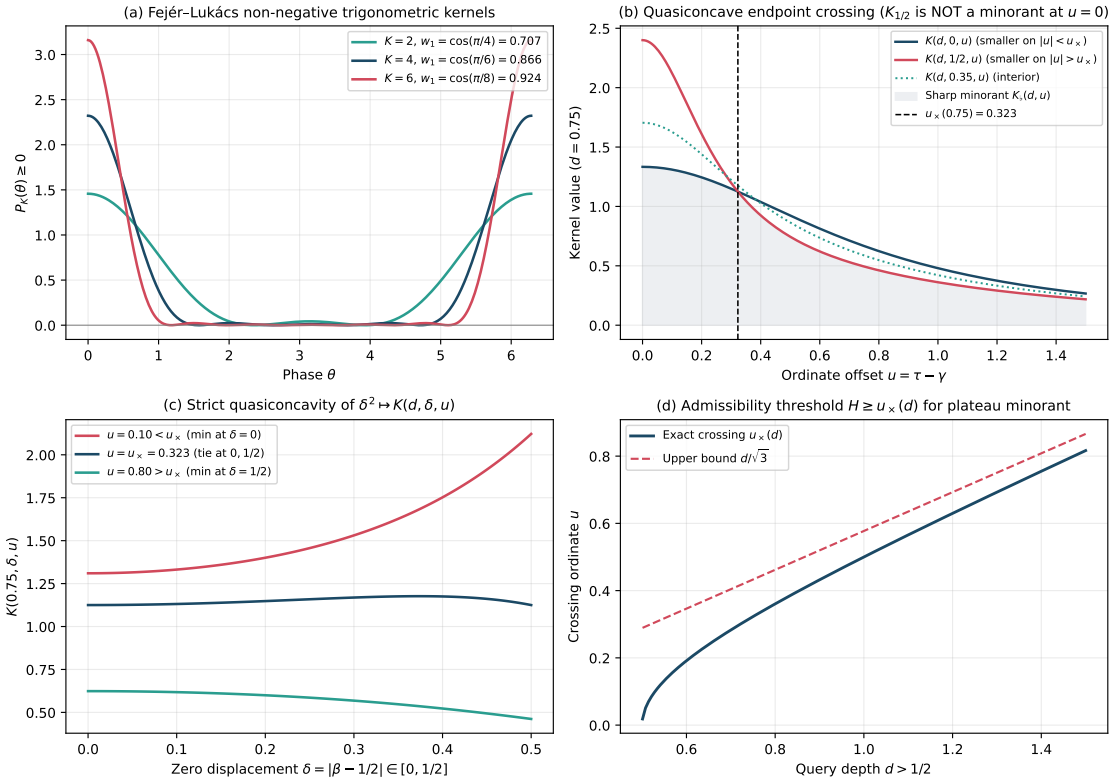


FIGURE 1. **(a)** Non-negative Fejér–Lukács kernels $P_K(\theta) \geq 0$. **(b)** Crossing of $K(d, 0, u)$ and $K(d, \frac{1}{2}, u)$ at $u_\times(0.75) \approx 0.323$ ($K(d, \frac{1}{2}, u)$ exceeds $K(d, 0, u)$ on $|u| < u_\times$). **(c)** Quasiconcavity of $\delta^2 \mapsto K(0.75, \delta, u)$ across $[0, \frac{1}{2}]$. **(d)** Crossing threshold $u_\times(d) < d/\sqrt{3}$.

Then for any putative off-line pair $\mathcal{Z}_0 = \{\rho_0, 1 - \bar{\rho}_0\}$ at $\gamma = t$ (setting $\mathbf{1}_{\text{focal}} = 1$ if $\tau = t$ and 0 if $\mathcal{Z}_0 = \varnothing$ or $\tau \neq t$),

$$\sum_{\rho \notin \mathcal{Z}_0} K(d, d_\rho, \tau - \gamma) \geq \mathcal{B}^{\text{rig}}_{\text{bg}}(d, \tau; H)$$

$$:= \left(\frac{1}{2\pi} \ln \frac{\tau}{2\pi} - \frac{1}{6\pi\tau^2}\right) \Phi_{\text{plat}}(d, H; M) - \frac{2}{3\pi\tau^2} I_2(d, H; M) \tag{5.2}$$

$$- K_{\min}(d, \tau - T_0)|S(T_0)| - B_S^{(0)}(\tau - H)\big[K_{\min}(d, H) - K_{\min}(d, \tau - T_0)\big]$$

$$- \big[B_S^{(0)}(\tau + H) + 2\mathbf{1}_{\text{focal}}\big] K_{\min}(d, H) - \frac{c_{\text{tail}}(\tau + H)}{\tau + H} \left(\frac{\pi}{2} - \Psi_{\text{win}}(d, H)\right).$$

Proof. Since $K(d, d_\rho, u) \geq \widetilde{K}_H(d, u) > 0$ for all $u \in \mathbb{R}$ by Theorem 4.1(iii) and all zeros have $|\gamma| > 14$, dropping $\gamma < 0$ gives $\sum_{\rho \notin \mathcal{Z}_0} K(d, d_\rho, \tau - \gamma) \geq \int_{T_0}^{\infty} \widetilde{K}_H(d, \tau - \gamma)\, \mathrm{d}N(\gamma) - 2\widetilde{K}_H(d, 0)\mathbf{1}_{\text{focal}}$. Using Backlund's identity $N(\gamma) = \frac{1}{\pi}\vartheta(\gamma) + 1 + S(\gamma)$ with $N(T_0) = 0$:

(1) *Smooth density integral.* Restricting $\int_{T_0}^{\infty} \widetilde{K}_H \frac{\vartheta'(\gamma)}{\pi}\, \mathrm{d}\gamma$ to $[\tau - M, \tau + M] \subset [15, \infty)$ and pairing $\tau \pm u$ ($u \in [0, M]$) with Stirling's lower bound $\vartheta'(x) \geq \frac{1}{2} \ln \frac{x}{2\pi} - \frac{1}{24x^2} \geq \frac{1}{2} \ln \frac{x}{2\pi} - \frac{1}{6\tau^2}$ ($x \geq \frac{\tau}{2}$) and $\frac{1}{2}[\ln(\tau + u) + \ln(\tau - u)] = \ln \tau + \frac{1}{2} \ln(1 - \frac{u^2}{\tau^2}) \geq \ln \tau - \frac{2u^2}{3\tau^2}$ (since $\ln(1 - w) \geq -\frac{4}{3}w$ on $[0, \frac{1}{4}]$) yields the first two terms of (5.2).

TABLE 1. Zero-free region constant $R_{\mathrm{eff}}(t)$ ($\sigma \geq 1-(R_{\mathrm{eff}}\ln t)^{-1}$) for the $K=4$ Fejér–Lukács polynomial without background subtraction vs. with the rigorous plateau bound (5.2), compared against Mossinghoff–Trudgian–Yang [13] ($R = 5.558691$) and the equivalent Korobov–Vinogradov factor.

Height t	$R_{\mathrm{Stechkin}}(t)$ ($K=4$, no bg)	$R_{\mathrm{PlatRig}}(t)$ (Theorem 5.1)	R_{MTY} [13]	$R_{\mathrm{KV,equiv}}(t)$ [13]
10^6	14.7628	12.8453	5.558691	31.7609
10^{12}	16.5113	13.5060	5.558691	27.2560
10^{20}	17.2811	13.4718	5.558691	24.1122
10^{40}	17.8964	13.2316	5.558691	20.2291

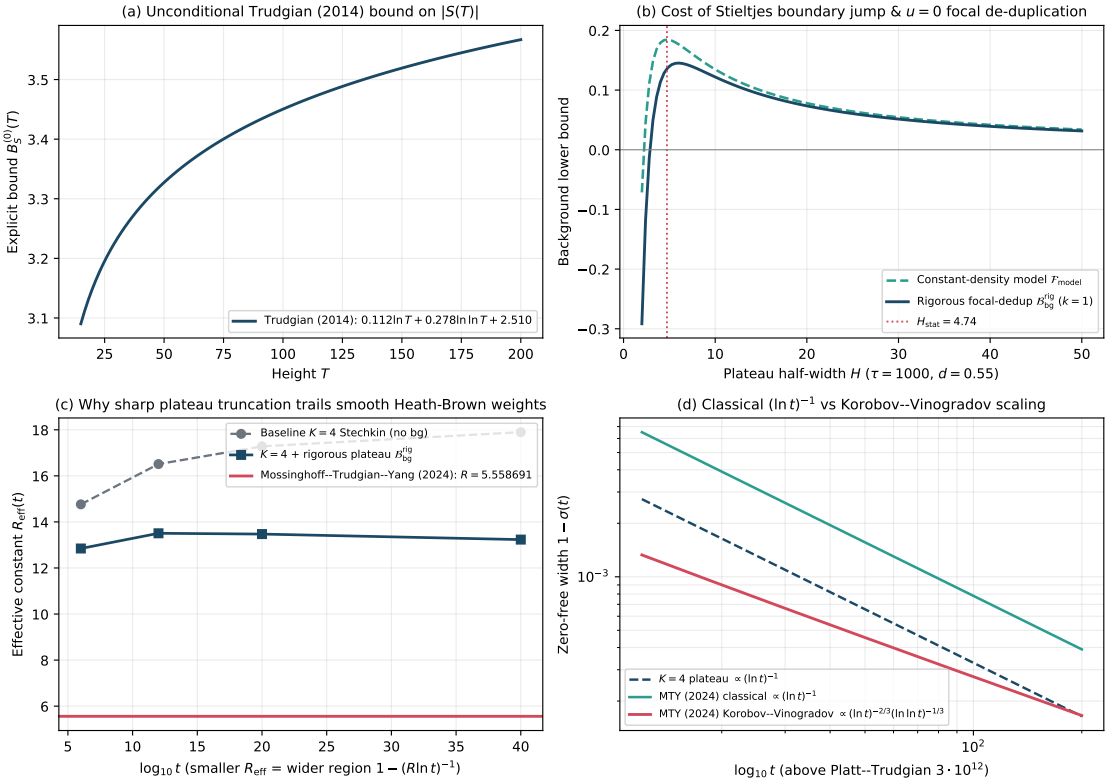


FIGURE 2. **(a)** Unconditional Trudgian (2014) bound $B_S^{(0)}(T)$ with constant 2.510. **(b)** Constant-density model $\mathcal{F}_{\mathrm{model}}$ vs. rigorous focal-deduplicated bound $\mathcal{B}_{\mathrm{bg}}^{\mathrm{rig}}$ at $\tau = 1000$. **(c)** Effective constant $R_{\mathrm{eff}}(t)$ compared with Mossinghoff–Trudgian–Yang ($R = 5.558691$). **(d)** Classical $(\ln t)^{-1}$ zero-free width vs. Korobov–Vinogradov scaling.

(2) *Stieltjes remainder.* Integrating $\int_{T_0}^{\infty} \widetilde{K}_H \, \mathrm{d}S$ by parts ($\partial_\gamma \widetilde{K}_H \equiv 0$ on $(\tau-H, \tau+H)$, >0 on $[T_0, \tau-H)$, and <0 on $(\tau+H, \infty)$) bounds the left segment by $B_S^{(0)}(\tau-H)[K_{\min}(d,H) - K_{\min}(d, \tau-T_0)]$. On $[\tau+H, \infty)$, concavity of $\ln x$ and $\ln\ln x$ gives $B_S^{(0)}(\tau+u) \leq B_S^{(0)}(\tau+H) + c_{\mathrm{tail}}(\tau+H)\ln(\frac{\tau+u}{\tau+H})$ for $u \geq H$; integrating $\ln(\frac{\tau+u}{\tau+H})(-\partial_u K_{\min}(d,u))$ by parts yields $\int_H^{\infty} \frac{K_{\min}(d,u)}{\tau+u}\,\mathrm{d}u \leq \frac{1}{\tau+H}(\frac{\pi}{2} - \Psi_{\mathrm{win}}(d,H))$. □

6. Scope, Boundary Conditions, and Open Questions

We summarize the methodological lessons established in Sections 2–5:

(1) *Why sharp/plateau Poisson truncation trails smooth Heath-Brown weights.* As Table 1 certifies, while $\mathcal{B}_{\mathrm{bg}}^{\mathrm{rig}}$ improves upon the baseline $K = 4$ Fejér–Toeplitz bound (reducing R_{eff} from 17.90 to 13.23 at $t = 10^{40}$), it remains substantially weaker than Mossinghoff–Trudgian–Yang's $R = 5.558691$ [13]. Two structural mechanisms explain this gap: first, $K_{\min}(d,u)$ has $\mathcal{O}(u^{-2})$ Lorentzian tails and slope discontinuities at $|u| = H$, incurring Stieltjes boundary penalties $\approx 2B_S^{(0)}(\tau)K_{\min}(d,H)$ plus focal de-duplication $-2K_{\min}(d,H)$; second, Heath-Brown's smooth compactly supported weight functions [4, 13] replace the raw Poisson kernel by rapidly decaying Mellin–Weil kernels whose derivatives are small everywhere, avoiding boundary jump penalties while jointly optimizing higher-degree cosine polynomials.

(2) *Asymptotic $(\ln t)^{-1}$ ceiling vs. Korobov–Vinogradov.* Every bounded-degree trigonometric inequality with $\sigma - 1 \asymp (\ln t)^{-1}$ yields at best a classical de la Vallée Poussin width $\mathcal{O}((\ln t)^{-1})$; reaching $(\ln t)^{-2/3}(\ln\ln t)^{-1/3}$ requires Vinogradov–Korobov exponential-sum bounds on $\zeta(s)$ [9, 21, 13]. Moreover, for $|t| \le 3\cdot 10^{12}$, all zeros already lie unconditionally on $\sigma = \frac{1}{2}$ by Platt–Trudgian [14].

Appendix A. Numerical and Interval Certification Protocols

In the standalone suite `scripts/verify_paper7_mirror_hadamard_variation.py`, all identities and bounds are verified in 50-decimal arithmetic (`mpmath.mp.dps = 50`) and 35-digit interval arithmetic (`mpmath.iv.dps = 35`):

(1) *Quasiconcave crossing certification.* Across $d \in \{0.55, 0.75, 1.00, 1.50\}$, $|K(d,0,u_\times) - K(d,\frac{1}{2},u_\times)| < 10^{-47}$, $K(d,0,0) - K(d,\frac{1}{2},0) < 0$, and interval arithmetic encloses $u_\times(0.60) \in [0.19148542, 0.19148543] < 0.60/\sqrt{3}$.

(2) *Validation against exact Riemann zeros.* Summing $K(d,0,\tau \mp \gamma_n)$ over the first 220 critical-line ordinates strictly exceeds $\mathcal{B}_{\mathrm{bg}}^{\mathrm{rig}}(d,\tau;H)$ at all test points (d,τ,H).

Use of AI and computational verification

In accordance with editorial policies on transparency in scientific computing, the author discloses that the AI research assistant **Gemini** (Google DeepMind) assisted in uncovering the $|u| < u_\times(d)$ crossing of $K(d,\frac{1}{2},u)$, proving quasiconcavity of $\delta^2 \mapsto K(d,\delta,u)$, correcting the Stieltjes tail, concavity, and focal de-duplication terms, and implementing `scripts/verify_paper7_mirror_hadamard_variation.py` (exit code 0).

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Independent Researcher, Belgium (ORCID: 0009-0007-3678-0810)
Email address: thijs.missiaen@gmail.com